

1. A uniform cylindrical coil in vacuum has radius  $r_1 = 1\text{ m}$ , length  $l_1 = 1\text{ m}$  and 100 turns. Coaxial and at the centre of this coil is a smaller coil of  $r_2 = 10\text{ cm}$ ,  $l_2 = 10\text{ cm}$  and 10 turns. Calculate the mutual inductance of the two coils.
2. The magnetic field in vacuum is given by  $\vec{B} = \hat{x}B_0 \cos(2x)\cos(\omega t - \beta y)$ . Find the displacement current, displacement vector and the volume charge density associated with this displacement vector.
3. In a **perfect conductor**, the conductivity is infinite, so  $\mathbf{E} = 0$  and any net charge resides on the surface.
  - (a) Show that the magnetic field is constant inside a perfect conductor.
  - (b) Show that the magnetic flux through a perfectly conducting loop is constant.
  - (c) **A superconductor** is a perfect conductor with additional property that the (constant)  $\mathbf{B}$  inside is in fact zero. (This flux exclusion is known as the Meissner effect). Show that the current in a superconductor is confined to the surface.
  - (d) Superconductivity is lost above a certain critical temperature ( $T_C$ ), which varies from one material to another. Suppose you had a sphere (radius  $a$ ) above its critical temperature and you held it in a uniform magnetic field  $\hat{z} B_0$  while cooling it below  $T_C$ . Find the induced surface current density  $\mathbf{K}$ , as a function of the polar angle  $\theta$ .
4. A very long solenoid of  $n$  turns per unit length carries a current which increases uniformly with time,  $i=kt$ ,  $k$  is a constant.
  - (a) Calculate the magnetic field inside the solenoid at time  $t$ .
  - (b) Calculate the electric field inside the solenoid.
  - (c) Consider a cylinder of length  $l$  and radius equal to that of the solenoid and coaxial with the solenoid. Find the rate at which energy flows into the volume enclosed by this cylinder and show that it is equal to  $\frac{d}{dt}\left(\frac{1}{2}LI^2\right)$ , where  $L$  is the self-inductance of the solenoid per unit length of the solenoid.
5. The electric field associated with a light wave propagating in vacuum is given by  $\vec{E} = \hat{x} 50e^{i(kz - 10^{15}t)}\text{ V/m}$   
Determine the direction of propagation, wave number, frequency and magnetic field component of the wave.
6. A plane wave in a vacuum has an electric field given by.  $\vec{E} = \hat{x} E_0 \cos[(kz - \omega t + \varphi)]$ 
  - (a) What is  $\vec{B}$ ?
  - (b) What is the complex amplitude of  $\vec{B}$ ?
  - (c) What are  $\vec{S}$  and  $\langle \vec{S} \rangle$ ?